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Supply chain performance measurement: a literature review

Goknur Arzu Akyuz* and Turan Erman Erkan

Department of Industrial Engineering, Atilim University, Kizilcasar Mahallesi, 06836 Incek Gölbasi, Ankara, Turkey

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This paper is intended to provide a critical literature review on supply chain performance measurement. The study aims at revealing the basic research methodologies/approaches followed, problem areas and requirements for the performance management of the new supply chain era. The review study covers articles coming from major journals related with the topic, including a taxonomy study and detailed investigation as to the methodologies, approaches and findings of these works. The methodology followed during the conduct of this research includes starting with a broad base of articles lying at the intersection of supply chain, information technology (IT), performance measurement and business process management topics and then screening the list to have a focus on supply chain performance measurement. Findings reveal that performance measurement in the new supply era is still an open area of research. Further need of research is identified regarding framework development, empirical cross-industry research and adoption of performance measurement systems for the requirements of the new era, to include the development of partnership, collaboration, agility, flexibility, information productivity and business excellence metrics. The contribution of this study lies in the taxonomy study, detailed description and treatment of methodologies followed and in shedding light on future research.

Keywords: supply chain; performance measurement; metrics; maturity

1. Introduction

Coordination of the supply chain (SC) has become strategically important as new forms of organisations, such as virtual enterprises, global manufacturing and logistics evolve. During the last few years, the focus has shifted from the factory level management of supply chains to enterprise level management of supply chains (Gunasekaran *et al*. 2005). Businesses becoming increasingly boundaryless (Puigjaner and Lainez 2008), increased challenges of globalisation, increased use of outsourcing, vendor managed inventory and advanced planning systems (APS), increased demands of integration led to a broadened supply chain definition (Meixell and Gargeya 2005). Differences between ‘traditional’ and ‘networked’ organisations are well discussed in Gunasekaran *et al*. (2005), emphasising the importance of strategic alliances, global outsourcing, shorter product life cycles, partnership formation and collaboration, agility, responsiveness, flexibility, reverse logistics and extended enterprise integration (integration beyond enterprise resources planning (ERP), covering both internal and external integration).

*Corresponding author. Email: arzuakyuz@atilim.edu.tr

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Integration, collaboration, and the use of IT are all depicted as 'building blocks' of 'house of supply chain' in Stadtler (2005). 'Increased importance of information systems' to support supply chain integration and management for the new organisation; and the idea that 'ERP provides the digital backbone in supply chain integration' are repeatedly emphasised in the literature (Pant *et al*. 2003, Bendoly and Kaefer 2004, Gunasekaran *et al*. 2004, Gunasekaran and Ngai 2004, Gunasekaran *et al*. 2005, Kelle and Akbulut 2005, Akyüz and Rehan 2009).

As such, recent technological developments in information systems and technologies have the potential to facilitate the coordination among different functions, allowing the virtual integration of the entire supply chain. The focus of this integration in the context of Internet-enabled activities is generally referred to as e-supply chain management (e-SCM), merging the two fields of supply chain management (SCM) and the Internet. e-SCM will refer to the impact that the Internet has on the integration of key business processes from end user to original suppliers that provide products, services and information that add value for customers and other stakeholders (Gimenez and Lourenço 2004).

With these trends in supply chain clearly proven, this paper aims at conducting a critical literature review to reveal the performance measurement requirements of today's broadened, e-enabled supply chains.

Essentiality of performance measurement in supply chain is vital, and Gunasekaran and Kobu (2007) mention the following as the purposes of a performance measurement system:

- Identifying success.
- Identifying if customer needs are met.
- Better understanding of processes.
- Identifying bottlenecks, waste, problems and improvement opportunities.
- Providing factual decisions.
- Enabling progress.
- Tracking progress.
- Facilitating a more open and transparent communication and co-operation.

Performance measurement is 'vital in strategy formulation and communication and in forming diagnostic control mechanisms by measuring actual results' (Wouters 2009).

The rest of the paper is organised as follows: Section 2 describes the review methodology, Section 3 mentions the basic characteristics and contributions of the works reviewed and Section 4 contains discussion and findings. Section 5 concludes and suggests future research directions.

2. Review methodology

The initial reading list for the review covered 42 articles from major science-cited journals. Because of the multi-disciplinary nature of the supply chain performance management topic, the papers which are located at the intersection of supply chain, IT, performance measurement and business process management were in the list to be able to provide a broad perspective covering technology, process and people's aspects. A taxonomy of these papers has been made and 24 papers are found much more relevant for the intersection of supply chain and performance measurement topics. As such, the review in this study is

Table 1. Distribution of the articles with respect to journals.

Computers and Chemical Engineering	2
Decision Support Systems	1
European Management Journal	1
International Journal of Production Economics	5
International Journal of Production Research	2
International Journal of Production and Operations Research	1
International Journal of Information Management	1
Journal of Modern Accounting and Auditing	1
Journal of Purchasing and Supply Management	1
Journal of Operations Management	1
Organisation Development Journal	1
Production Planning & Control	1
Supply Chain Management: An International Journal	2
Technovation	1
Total Quality Management	1
Transportation Research	2
Total	24

based on 24 papers from major journals. Distribution of these articles with respect to journals is given in Table 1.

The list of papers included in review and their classification with respect to their topic and methodology are given in Table 2.

Focus, contributions and approaches are summarised in Table 3.

As can be seen in Tables 2 and 3, review style papers and questionnaire-based surveys using statistical techniques for the analysis dominate the selected list. Some of the papers in the list use more rigorous approaches: Perea *et al*. (2000) use dynamic modelling combined with classical control theory; Puigjaner and Liainez (2008) utilise a multi-stage, stochastic mixed integer linear model to capture the supply chain dynamics; Cai *et al*. (2008) suggest an iterative analytical approach based on eigenvalues for dependance modelling of key performance indicators (KPIs); and Hwang *et al*. (2008) use stepwise regression to analyse dependancy of measures. Bhagwat and Sharma (2007) use the analytical hierarchy process (AHP) approach for prioritising metrics. Ho (2007) adopts a simulation-based, experimental approach for ERP-based supply chain system performance measurement.

The taxonomy matrix (topic versus methodology) for the reviewed papers is given in Table 4.

3. Basic characteristics and contribution of the works reviewed

The papers included in the review can be categorised into six main subgroups according to their common themes:

(i) General trends and issues in supply chain.
(ii) Dynamic modelling approaches.
(iii) Supply chain performance management issues.
(iv) Process maturity-supply chain performance relation.
(v) KPI prioritisation and dependence.
(vi) Human and organisational sides of supply chain performance management.

Table 2. Topic and methodology classification.

				Topic								Methodology			
No.	Author	Year	Title	ERP	IT	Supplychain	E – supplychain	BPM	Technology/BPMfit	Performance measurement/ metrics	Roadmap implementation success	Review	Case – based/ survey	Conceptual model/ framework	Mathapproaches
1	Meixell, M.J. and Gargeya, V.B.	2005	Global supply chain design			√						√			
2	Vonderembrese, M.A. *et al.*	2006	Designing supply chains: towards theory development			√						√	√		
3	Swofford, P. *et al.*	2008	Achieving supply chain agility through IT integration and flexibility			√			√	√			√	√	
4	Puiganer, L. and Lainez, J.M.	2008	Capturing dynamics in integrated SCM			√									√
5	Perea, E. *et al.*	2000	Dynamic modeling and classical control theory for SCM			√									√
6	Gunasekaran, A. *et al.*	2004	A framework for supply chain performance measurement			√				√			√	√	
7	Martin, P.R. and Patterson, J.W.	2009	On measuring company performance within a supply chain			√				√			√		
8	Gunesekaran, A. and Kobu, B.	2007	Performance measures and metrics: a review of recent literature			√				√		√			
9	Gunesekaran, A. *et al.*	2005	Performance measurement and costing system in new enterprise							√				√	
10	Yao, K. and Liu, C.	2006	An integrated approach for measuring supply chain performance							√				√	
11	Ho, C.	2007	Measuring system performance of an ERP-based supply chain	√		√				√			√		√

12	Bernardes, E. and Zsidisin, G.	2008	An examination of strategic supply management benefits and performance implications				√			√	√	
13	Lockamy, L. and McCormack, K.	2004	Linking SCOR planning practices to supply chain performance	√			√	√		√	√	
14	McCormack, K. and Lockamy, L.	2004	The development of a supply chain management process maturity model using concepts of business process orientation	√			√	√		√	√	
15	McCormack, K. *et al.*	2008	Supply chain maturity and performance in Brazil	√			√	√	√	√	√	
16	Baghwat, R. and Sharma, M.K.	2007	Performance measurement of supply chain management using the hierarchical process	√			√			√		√
17	Cai, J. *et al.*	2008	Improving supply chain performance management: a systemic approach to analysing iterative KPI accomplishment				√			√	√	√
18	Hwang, Y. *et al.*	2008	The performance evaluation of SCOR sourcing process				√			√		√
19	Kanji, G. and Wong, A.	1999	Business excellence model for supply chain management	√	√		√			√	√	
20	Robinson, J.R. and Malhotra, M.K.	2005	Defining the supply chain quality management and its relevance to academic and industrial practice	√	√		√		√	√		
21	Wouters, M.	2009	A developmental approach to performance measures: results from a longitudinal case study				√			√		
22	Stock, G. *et al.*	2000	Enterprise logistics and supply chain structure: role of fit	√		√				√	√	
23	Geiger, S.	2006	Strategy/structure fit and firm performance			√				√	√	
24	Butterman, G. *et al.*	2008	Contingency theory 'fit' as gestalt: an application to supply chain management			√				√		

Table 3. Classification with respect to focus and contribution.

No.	Author	Year	Title	Focus	Contribution/approach
1	Meixell, M.J. and Gargeya, V.B.	2005	Global supply chain design	Emerging issues in global SC	Comprehensive review and classification. Critiques emerging trends in historical perspective. Emphasises outsourcing, VMI, integration across tiers, internal and external integration, and performance measurement criteria.
2	Vonderembrese, M.A. *et al.*	2006	Designing supply chains: Towards theory development	Product life cycle supply chain types matching, including agility and lean classifications	Detailed descriptions of lean and agile SC, tries to match product life cycles and product types with different supply chain types. Supports with three cases.
3	Swofford, P. *et al.*	2008	Achieving supply chain agility through IT integration and flexibility	Relationship among IT integration, SC flexibility, SC agility and business performance	Tests the relationships of IT integration, SC flexibility, SC agility and competitive business performance.
4	Puiganer, L. and Lainez, J.M.	2008	Capturing dynamics in integrated SCM	Dynamic behaviour modelling	Multi-stage, multi-period, stochastic mixed integer linear model combined with control theory. Develops a strategic-level model, uses forecasting, optimisation and simulation in tandem, analyses results using sample scenarios. The model involves demand and price uncertainty, financials (assets, liabilities, credit policies, capacity expansion, Shareholder value).
5	Perea, E. *et al.*	2000	Dynamic modeling and classical control theory for SCM	SC modelling with dynamic modelling	Development of a dynamic model involving laws and state transitions.
6	Gunasekaran, A. *et al.*	2004	A framework for supply chain performance measurement	Performance measurement	Measurement and metrics classification. Involves survey. Assessing importance for each performance measure.
7	Martin, P.R. and Patterson, J.W.	2009	On measuring company performance within a supply chain	Identification of different performance measures	Defines three main classes of performance measures: inventory, cycle time and financials. Uses a survey to investigate the effects of supply relations organisational structure, partnering, supplier agreements and process improvements.

8	Gunesekaran, A. and Kobu, B.	2007	Performance measures and metrics: a review of recent literature	SC performance measurement	Comprehensive review and classification. Justification for the need of new metrics to support new organisations. Need and purpose of performance measurement, criteria for successful metrics well discussed. Classification of different measurement perspectives.
9	Gunesekaran, A. *et al.*	2005	Performance measurement and costing system in new enterprise	Performance-based costing system for the new enterprise	Comprehensive discussion of pressures and approaches for the new organisation. Direct justification for the need of a new performance measurement and costing system. Development of a framework.
10	Yao, K. and Liu, C.	2006	An integrated approach for measuring supply chain performance	EVA, BSC and ABC in SC	Combines EVA, BSC, ABC. Suggests use of various KPIs and a framework.
11	Ho, C.	2007	Measuring system performance of an ERP-based supply chain	ERP-based supply chain performance	Proposes an integrated method, total related cost measurement, to evaluate supply chain performance of a 3-echelon, ERP-based supply chain system. Uses simulation-based validation experiments.
12	Bernardes, E. and Zsidisin, G.	2008	An examination of strategic supply management benefits and performance implications	Relation of strategic supply chain management with the concepts of network embeddedness and network scanning	Survey-based study focusing on network embeddedness and scanning. Rigorous statistical treatment.
13	Lockamy, L. and McCormack, K.	2004	Linking SCOR planning practices to supply chain performance	SCOR planning practice and supply chain performance relationships	Survey-based study to investigate relationship of SCOR planning practices and performance.
14	McCormack, K. and Lockamy, L.	2004	The development of a supply chain management process maturity model using concepts of business process orientation	Maturity model and performance relationship	Develops a maturity model having a business process view. Defines 5 levels of maturity and performs a survey to investigate the relationship of maturity and performance.
15	McCormack, K. *et al.*	2008	Supply chain maturity and performance in Brazil	Innovative performance measurement and maturity model	Takes the SCOR model and business process orientation maturity model as base. Develops a Brazilian survey. Provides clear support for new performance measurement and maturity model. Includes clear support for the development of new performance measurement methodologies and clearly emphasises the need and importance of survey-based studies.

(Continued)

Table 3. continued.

No.	Author	Year	Title	Focus	Contribution/approach
16	Baghwat, R. and Sharma, M.K.	2007	Performance measurement of supply chain management using the hierarchical process	Prioritisation and choice of metrics and measures	Proposes 5 classes of metrics and proposes an AHP approach. Supports with a survey. Comprehensive review of BSC and AHP.
17	Cai, J. *et al.*	2008	Improving supply chain performance management: a systemic approach to analysing iterative KPI accomplishment	Dependence and priority modelling of KPIs	Challenges, intricacy dependency and conflicts of performance measurement system. Iterative, analytical approach based on eigen values. Tries to model dependency on KPIs. Checks the cost of improving KPIs at each iteration.
18	Hwang, Y. *et al.*	2008	The performance evaluation of SCOR sourcing process	SCOR-based Taiwanian case study to evaluate sourcing	SCOR overview, Taiwanian LCD sector questionnaire, stepwise regression analysis to analyse dependency of measures and a rigorous statistical test and justification.
19	Kanji, G. and Wong, A.	1999	Business excellence model for supply chain management	Supply and business excellence	Develops an excellence model similar to EQQM. Verifies the model with a survey. Emphasis is on the concept of extended TQM and the need for excellence in all processes.
20	Robinson, J.R. and Malhotra, M.K.	2005	Defining the supply chain quality management and its relevance to academic and industrial practice	SC quality focus-extended quality	Defines SC quality management Merges quality and supply domain. Provides a very comprehensive taxonomy. Idea of supply chain excellence is emphasised. Provides clear support for overall performance measurement. Includes a survey-based study.

21	Wouters, M.	2009	A developmental approach to performance measures: results from a longitudinal case study	Concept of enabling performance management	Challenges of performance measurement, a company-based study, need of developmental approach in performance measurement, importance of delegating the performance measurement at every level of hierarchy. Emphasises the idea of 'metrics for people'.
22	Stock, G. *et al.*	2000	Enterprise logistics and supply chain structure: role of fit	Logistics and SC structure elements. Concept of fit	Review section comprehensive and develops a framework of fit between logistics integration and SC structure. Defines fit variables and analyses with a survey.
23	Geiger, S. *et al.*	2006	Strategy/structure fit and firm performance	Relationship between fit and performance	Emphasises the mediating effect of industry concentration between fit and performance. Contains manufacturing-based survey. Develops a relation to measure return on assets.
24	Butterman, G. *et al.*	2008	Contingency theory 'fit' as gestalt: an application to supply chain management	Fit of strategy, structure and IT	Survey-based clustering analysis for fit of strategy, structure and IT variables. Ends up with 6 levels of maturity. Clustering levels can be a base for our study. A critical application of theory of 'fit' to supply chain.

Table 4. Taxonomy matrix.

	ERP	Supply chain	BPM	Technology/ BPM fit	Performance measurement/ metrics	Roadmap/ implementation success	Turkish implementation
Review		1, 2, 8, 15, 20	20		8, 15, 20	15	
Case-based/ survey	11	2, 3, 6, 7, 11, 13, 14, 15, 16, 19, 20, 22	19.2	3, 22, 23, 24	3, 6, 7, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21	13, 14, 15	
Model/ framework		3, 6, 13, 14, 15, 19, 22	19	3, 22, 23	3, 6, 9, 10, 14, 15, 17, 18, 19	14, 15	
Math approaches	11	4, 5, 11, 16			11, 16, 17, 18		

In this section, characteristics and contributions of the works reviewed are discussed in detail under these six subheadings.

3.1 *Papers that focus on 'general trends and issues in supply chain'*

Meixell and Gargeya (2005) provide a comprehensive, critical review and classification of global supply chain literature and put forward the emerging trends in historical perspective. Outsourcing, vendor managed inventory (VMI), integration across tiers, internal and external integration, and the need of various performance measurement criteria are emphasised as the main trends.

A matching between product life cycle and types of supply chain, including agility and lean supply chain classifications, is suggested by Vonderembrese *et al*. (2006). Detailed descriptions of lean and agile supply chain are provided and their work is supported with three case studies: Black & Decker, IBM and Daimler Chrysler.

Swafford *et al*. (2008) investigate the relationship among IT integration, SC flexibility, SC agility and business performance through a US case-based study. Their study reveals the 'domino effect' among IT integration, SC flexibility, SC agility and competitive business performance.

This group of papers clearly reveal the main trends and the importance of the IT integration, flexibility, agility and lean concepts for today's supply chain management.

3.2 *Papers using 'dynamic modelling' approach*

Puigyaner and Lainez (2008) use multi-stage, multi-period, stochastic mixed integer linear model combined with control theory to optimise corporate value. They develop a strategic-level model using forecasting, optimisation and simulation in tandem, and analyse the results using sample scenarios. Their comprehensive model involves demand and price uncertainity and financials (assets, liabilities, credit policies, capacity expansion, shareholder value, etc).

Perea *et al.* (2008) use dynamic modelling approach combined with classical control theory to develop a generic dynamic framework for supply chain modelling.

These two papers emphasise the importance of capturing supply chain dynamics at various decision levels and they are clear indications that modelling efforts to handle these dynamics are still continuing in literature.

3.3 *Papers having direct focus on 'supply chain performance management'*

Papers categorised in this group deal with various aspects of performance measurement system, including metrics classifications, problems of the current performance measurement systems and the need for the establishment of a new performance measurement.

Gunasekaran *et al.* (2004) develop a framework for supply chain performance measurement. The article provides a detailed 'measurement and metrics classification' and uses a survey aiming at assessing importance within each metric group.

Three main classes of performance measures are discussed by Martin and Patterson (2009): inventory, cycle time and financials. Effects of supply relations (organisational structure, partnering, supplier agreements and process improvements) on the performance measures selected are investigated via a survey-based study.

Gunasekaran and Kobu (2007) offer a comprehensive review and classification for supply chain measurement and metrics. A trend of increasing attention on performance measurement and metrics, both in practice and literature, is emphasised in their work. This idea is also supported by McCormack *et al.* (2008). Gunasekaran and Kobu (2007) highlights the confusion as to the classification of metrics in literature, and lacking complete coverage of all the performance measures. Their review classifies the literature based on the following criteria: balanced scorecard perspective, components of measures, location of measures, decision levels, nature of measures, measurement base, traditional versus modern measures. They treat a number of metrics in five classes: order planning, supplier evaluation, production level, delivery and customer and they conduct an empirical research to assign importance ratings within each class. The work is a clear support for the need of new metrics for the new organisation.

A comprehensive discussion of pressures and approaches for the new organisation appears in Gunasekaran *et al.* (2005). The study is also the direct justification for the need of a new performance measurement and costing system.

Supporting the idea of new performance measurement system, Yao and Liu (2006) and Ho (2007) propose different approaches. Yao and Liu (2006) suggest an integrated approach for measuring supply chain performance, combining economic value added (EVA), the balanced scorecard (BSC) and activity based costing (ABC), clearly emphasising the need of overhead handling and a balanced approach. Ho (2007) focuses on ERP-based supply chain performance and proposes an integrated method, total related cost measurement, to evaluate supply chain performance of a three-echelon, ERP-based supply chain system. The study uses simulation-based validation experiments.

Bernardes and Zsidisin (2008) investigate the relation of strategic supply chain management with the concepts of network embeddedness and network scanning, specifically focusing on the concept of embeddedness and network scanning in relation to performance. Their work involves a survey-based study made in US manufacturing, supported by rigorous statistical analysis.

Papers under this subsection point to problems of the current performance measurement system and provide clear evidence that literature is still in need of a new supply chain performance measurement system which can handle the requirements of the new supply chain era.

3.4 *Papers investigating the process maturity–supply chain performance relation*

Three papers by McCormack focus on process maturity concept and investigate the relation with respect to supply chain performance.

Lockamy and McCormack (2004) investigate the relationship between supply chain management planning practices and supply chain performance based on four main decision areas of SCOR model (plan, source, make, deliver) and result in the importance of planning function and the importance of collaboration, process measures, process collaboration, process credibility, process integration and information technology.

McCormack and Lockamy (2004) develop a process maturity model taking the business orientation view, defining five general levels of process maturity and using the survey instrument to analyse the relationship of process maturity with performance.

McCormack *et al*. (2008) take the supply chain operations reference (SCOR) model and business process orientation maturity model of McCormack and Lockamy (2004) as a base. The study provides a comparison on the traditional versus innovative performance measurement systems. A Brazilian survey is conducted in the study for clustering performance of the companies surveyed. The study puts forward a clear support for the need of new performance measurement methodologies and maturity models, emphasising the importance of survey-based studies.

These three papers highlight the maturity and performance relationship and provide clear evidence that literature is still in search of maturity models and roadmaps, which are proven to have direct correlation with performance.

3.5 *Papers focusing on modelling, prioritisation and dependance modelling of KPIs*

Papers classsified in this group aim at dealing with hierarchical nature, dependancy and complexities of KPIs and suggest various approaches to handle these complexities.

Bhagwat and Sharma (2007) provide a comprehensive review on BSC and AHP, focusing on prioritisation and choice of metrics and measures. They propose an AHP approach based on a Western-India survey.

The challenge, intricacy, dependancy and conflicts of supply chain performance measurement system are emphasised by Cai *et al*. (2008). They utilise an iterative, analytical approach based on Eigen values and suggest a model to handle KPI dependancies, considering the cost of improving KPIs at each iteration.

Hwang *et al*. (2008) performed a case-based study for the Taiwanese TFT-LCD (thin film transistor-liquid crystal display) manufacturing sector. Their work contains a comprehensive SCOR overview and stepwise regression analysis to analyse the dependancy of different performance measures. They specifically focus on the ‘sourcing’ side of the SCOR model.

It is evident that modelling the hierarchical nature and dependancies among various KPI's is still an unresolved and challenging issue in supply chain domain.

3.6 *Papers focusing on the 'human/organisational' sides of the performance management*

Papers falling in this class deal with the concepts of 'enabling performance management', 'total supply chain quality' and the concept of 'fit' in relation to performance measurement.

Kanji and Wong (1999) point out the 'human side' of the issue is not covered in most of the work on SCM. The concept of total quality management (TQM) is extended to supply chain and the need for 'business excellence indices' is highlighted. In today's understanding of supply chain excellence, collaboration, agility and flexibility are among the critical success criteria and today's supply chain performance management still appears to be having difficulty in measuring the degree of collaboration, agility and flexibility.

Robinson and Malhotra (2005) focus on quality management requirements of the new supply chain era and mention supply chain quality management concept, emphasising the commitment to quality both inter- and intra-organisationally, again basing on the SCOR model and balanced scorecard approach. The paper provides a clear support for the need for further research in SC Quality management area.

Wouters (2009) mentions the concept of 'enabling performance management', emphasising the need for involvement of people at all levels, starting with the determination of the metrics. Challenges of performance measurement, need of developmental approach in performance measurement, importance of delegating the performance measurement at every level of hierarchy and the idea of 'metrics for people' are treated in detail. His previous work, Wouters and Wilderom (2008) is also referenced in this work and the study is critical in emphasising the need for longitudinal case studies.

Stock *et al.* (2000) define the concept of 'fit' as the appropriate consistency between logistics practices and supply chain structures and investigates the impact of fit among channel governance, geographical dispersion and logistics integration on supply chain performance. Their study provides support for the importance of 'fit' among various supply chain parameters.

Geiger *et al.* (2006) investigate the relationship of strategy/structure fit and firm performance using the mediating factor of 'industry concentration'. They reveal a clear need to analyse the effects of mediating factors other than industry concentration.

Buttermann *et al.* (2008) present an application of 'fit' as Gestalt perspective to supply chain management. Fit is mentioned as 'mediation, moderation, matching, covariation, profile deviation and gestalts'. Their study applies fit as Gestalt perpective to search for archetypes or 'recurrring clusters of attributes' which are directly related to the performance and the use of these archetypes as a means for classification of firm performance. Using a survey-based study, they identify six main archetypes: simple, low performers, market performers, average players, internally integrated low performers, masters of efficiency and two-time winners. It is emphasised that this is the first-time 'fit as gestalt concept' is applied to SCM.

This group of papers clearly indicate the need for having a broad, organisation-wide perspective of the issue, highlighting the importance of consistency among various organisational factors. It also became apparent that the issue of 'fit' deserves further attention.

4. Discussion and findings

This section includes discussion and findings under four subsections:

(i) Problems in today's PMS.
(ii) Requirements for performance measurement metrics.
(iii) Importance of 'balanced scorecard' approach and SCOR model.
(iv) Importance of 'concept of fit' in supply chain performance measurement.

4.1 *Problems in today's PMS*

This review clearly put forward the problems of today's performance measurement systems. In today's competitive age, it is proven that many companies have not succeeded in maximising their supply chain's potential because they have often failed to develop the performance measures and metrics needed to fully integrate their supply chain to maximise effectiveness and efficiency (Gunasekaran *et al*. 2004). The following are pointed out as the main problems in performance measurement by Gunasekaran *et al*. (2004) and Gunasekaran and Kobu (2007):

- Incompleteness and inconsistencies in performance measurement and metrics.
- Failing to represent a set of financial and non-financial measures in a balanced framework, some concentrating on financials, others concentrating on operational measures.
- Having a large number of metrics, making it difficult to identify the critical few among trivial many.
- Failing to connect the strategy and the measurement.
- Having a biased focus on financial metrics.
- Being too much inward looking.

With all these problems highlighted, there seems to be no universal consensus regarding suitable measures of supply chain quality performance, and commonly implemented supply chain measurements are fragmented and virtually unknown (Robinson and Malhotra 2005). Since many measurement systems lacked strategy alignment, a balanced approach and systemic thinking, they have difficulty in systematically identifying the most appropriate metrics (Cai *et al*. 2008). The work of Cai *et al*. (2008) also states that these measurement systems do not provide a definite cause–effect relationship among numerous and hierarchial individual KPIs. The fact that 'since many measurement systems are static, they lag the trend' is also mentioned. The importance of hierarchy and dependance among different KPIs are also highlighted in Hwang *et al*. (2008).

Gunasekaran *et al*. (2005) emphasise the need to handle predominant overheads accurately, while providing non-financial information and Gunasekaran and Kobu (2007) expressed the need and importance of using KPIs measuring 'innovation'.

McCormack *et al*. (2008) compare the traditional and innovative performance measurement (PMS) as given in Table 5, indicating the changes required over the traditional performance measurement systems.

This table clearly puts forward the importance of long term value orientation and compatibility among innovative requirements for today's performance measurement.

Work by Robinson and Malhotra (2005) and Wouters (2009) clearly supports the need for a performance measurement system taking the holistic picture, including the human side and organisational issues.

Table 5. Comparison of traditional vs. innovative PMS.

Traditional PMS	Innovative PMS
Based on cost/efficiency	Based on value
Trade-off between performances	Compatibility of performances
Profit oriented	Client oriented
Short term orientation	Long term orientation
Individual metrics prevail	Team metrics prevail
Functional metrics prevail	Transversal metrics prevail
Comparison with the standard	Monitoring of improvement
Aimed at evaluation	Aimed at evaluation and involvement

Source: McCormack *et al*. (2008).

The above literature items provide clear proof for the deficiencies of the current performance measurement systems and for the significant changes required over traditional performance measurement.

4.2 *Requirements for performance measurement metrics*

Taking into account the previous considerations and the comprehensive explanations of Gunasekaran *et al*. (2004), Gunasekaran and Kobu (2007) and Wauters (2009) on the basic characteristics and requirements of proper performance measurement and metrics, it is possible to argue that new era performance measurement metrics should:

- Truly capture the essence of organisational performance.
- Base on company strategy and objectives.
- Reflect a balance between financial and non-financial measures.
- Relate to strategic, tactical and operational levels of decision making and control.
- Be comparable to other performance measures used by similar organisations.
- Clearly define the purpose, data collection and calculation methods, update and monitoring mechanisms and related procedures.
- Vary between organisational locations and be under control of the evalated organisational unit.
- Allow for setting targets, aggregation and disaggregation.
- Allow prioritisation/weighting.
- Facilitate integration.
- Avoid overlaps.
- Be able to handle complex overhead structures.
- Be simple and easy to use, preferably in the form of ratios rather than absolute numbers.
- Be specific and non-financial, rather than aggregate and financial, to be more actionable.
- Be determined through discussion with all the parties involved and serve the needs of people from all levels (not only upper management).
- Adopt a proactive approach, enabling fast feedback and continuous improvement.
- Be valid and reliable.

- Be coherent and transparent.
- Be experience based.
- Allow for testing, reviewing, revising and refining, which involves organisational learning.
- Result in minimum number of indicators that provide reasonable accuracy with minimum cost.
- Be able to measure partnership, collaboration, agility, flexibility, information productivity and be able to define business excellence.

It is evident that establishing and implementing a performance measurement system to meet all these requirements is a challenging task requiring simultaneous considerations of business process management, technical and organisational/managerial issues. These challenges are amplified by increased pressures for measuring partnership, collaboration, agility, and business excellence requirements of the new era. As such, the issue requires a 'balanced', 'organisation-wide', 'dynamic' and 'continuous learning' approach based on sound business process management practices.

4.3 *Importance of 'balanced scorecard' approach and SCOR model*

Balanced scorecard methodology by Kaplan and Norton (1993, 1996), rooted to their 1993 work, still lies at the heart of today's performance management system. Current literature reveals that the need and importance of balanced scorecard approach for today's supply chain performance measurement is definitely beyond discussion. The idea of hierarchial, balanced set of performance metrics compatible with the top management strategy is repeatedly emphasised and lies at the heart of requirements of a performance measurement system. An overall balance is sought for between:

- Short term vs. long term.
- Internal vs. external focus.
- Different levels in an organisation.
- Four views of BSC (learning and growth, internal processes, customer, financials).
- Multiple perspectives of stakeholders (Bhagwat and Sharma 2007).

Importance of measurements related with intangible assets (human, information and organisational capital) is also evident in today's balanced scorecard perspective, as also emphasised in Kaplan and Norton (2004).

Literature also reveals that with the recent developments, the SCOR model created by the SCC (Supply Chain Council) gained growing use and increased visibility, contributing to the development and evolution of supply chain performance measurement systems and maturity models by:

- Providing a standardised way of viewing the supply chain (cross-industry standard).
- Offering a consistent 'scorecard' framework for development of performance.
- Emphasising process orientation and deemphasising functional orientation.
- Enabling cross-industry benchmarks.

Lockamy and McCormack (2004), Cai *et al*. (2008), Hwang *et al*. (2008) and McCormack *et al*. (2008) all clearly support the importance of the SCOR model as a base in current SC performance measurement.

4.4 *Importance of 'concept of fit' in supply chain performance measurement*

Besides the idea of alignment of 'strategy' and 'performance measurement and metrics', there is significant evidence in literature as to the importance of the concept of 'fit' in supply chain literature. This review suggest that the idea of 'fit' among various parameters has direct performance implications for supply chain and application of these ideas to supply chain is still immature in literature. Case-based studies to analyse the effects of 'degree of fit' among various parameters on different performance measures are still worth investigating. Finding out the effects of various 'mediating' variables on 'fit-performance' relationship and developing generic models/paths of maturity are topics that still deserve further attention in the supply chain domain.

5. Conclusions and future research directions

This study has put forward the problems and requirements of today's broadened, e-enabled supply chain performance measurements systems as distinctive from the traditional performance measurement systems. The importance of the 'balanced scorecard' approach and significance of the SCOR model as the foundation of the performance management system are highlighted during the study. Multidimensional nature of the issue is evident, involving the concepts of 'total quality', 'fit' and 'excellence'.

The study revealed that supply chain performance measurement is still a fruitful research area and very distinctive supportive statements have been traced for the need of further research on supply chain performance measurement during the review. The following are the main guidelines identified for future reseach:

- More research on the performance measurement tools for 21st century business models, need for the development of more precise frameworks and empirical testing of the performance measures, action research.
- Validation of developed performance measures, determination of KPI's for partnership; and development of models to cover virtual and e-commerce environments.
- Developing measurement and performance systems in the form of new maturity models supported by SCOR, to enable benchmarking.
- Need for cross-industry studies.
- Need for development of metrics for measuring the performance and suitability of IT in SCM.
- Performance measurement and metrics for responsive SC.

Immaturity of the frameworks and models are evident in this survey and the authors believe that future contributions to the area will come specifically from:

- Framework development efforts.
- Development of partnership, collaboration, agility, flexibility, information productivity and business excellence metrics.
- Further elaboration on the fit–performance relationships, including modelling and case-based surveys.

The authors believe that 'total quality', 'business process', 'fit' and 'excellence' ideas are still the key for performance measurement systems of future. The survey provided strong support as to the immaturity of these concepts in relation to supply chain. To put it clearly, 'supply chain business excellence' deserves further attention in any future research.

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